

COTS Fine-Scale SDM: Comprehensive Validation Report

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Overview

This report validates the XGBoost COTS Fine-Scale Species Distribution Model using historical Manta Tow and Cull data. We compare both the **2025 Year-Specific** and **Year-Agnostic** models.

The validation involves: 1. **Grid-Based Historical Validation (100m)**: General statistics (ROC, AUC, Confusion Matrix) evaluating the model against historical COTS/scar presence.

2. **Tow-Path Simulation:** Operational simulation showing which manta tows would be suggested based on the model. 3. **Cull Dive Financial Savings:** Quantification of dive hours and dollars saved by avoiding ineffective culls (0.00 or <0.02 CPUE). 4. **Reef Maps:** Visual mapping of predictions vs. operations.

```
library(terra)
library(sf)
library(dplyr)
library(readr)
library(readxl)
library(ggplot2)
library(tidyr)
library(knitr)
library(stringr)
library(pROC)

# --- Configuration ---
predict_agnostic_path <- "outputs/COTS_prob_0.02_cpue_agnostic.tif"
predict_year2025_path <- "outputs/COTS_prob_0.02_cpue_year2025.tif" # native 30m raster
manta_csv <- "data/COTS Program Manta Tow Data-2026-02-04.csv"
cull_xlsx <- "data/250929_COTS-Manta-Cull-RHIS-Data-Matthews-and-Schlawinsky.xlsx"
cull_polys_gpkg <- "data/cull_site_polygons.gpkg"

out_dir <- "validation_outputs"
dir.create(out_dir, showWarnings = FALSE, recursive = TRUE)

# Selected Reefs
reef_select <- c(
  "Tongue Reef (16-026)",
  "Batt Reef (16-029)",
  "Arlington Reef (16-064)",
  "(Big) Broadhurst Reef (No 1) (18-100a)",
  "Britomart Reef (18-024)",
  "Sudbury Reef (No 1) (17-001a)",
  "Trunk Reef (18-027)",
  "Bramble Reef (18-029)",
  "Davies Reef (18-096)",
  "Heron Reef (23-052a)"
)
```

1. Load and Prepare Data

```
# Load Model Predictions
prob_agnostic <- if(file.exists(predict_agnostic_path)) terra::rast(predict_agnostic_path) e
prob_year2025 <- if(file.exists(predict_year2025_path)) terra::rast(predict_year2025_path) e

if(is.null(prob_year2025) || is.null(prob_agnostic)) stop("Missing prediction rasters.")

# Load Manta Tow Data
manta_raw <- read_csv(manta_csv, show_col_types = FALSE) %>%
  filter(
    !is.na(StartLongitude), !is.na(StartLatitude), !is.na(EndLongitude), !is.na(EndLatitude)
    StartLongitude != 0, StartLatitude != 0, EndLongitude != 0, EndLatitude != 0
  ) %>%
  mutate(
    cots_observed = CrownOfThornsStarfishCount > 0,
    scars_observed = !is.na(ScarsCount) & ScarsCount > 0,
    presence_any = cots_observed | scars_observed,
    ReefName = as.character(ReefName)
  )

# Convert Manta Tows to spatial lines
make_tow_lines <- function(df) {
  lines <- lapply(seq_len(nrow(df)), function(i) {
    p1 <- c(df$StartLongitude[i], df$StartLatitude[i])
    p2 <- c(df$EndLongitude[i], df$EndLatitude[i])
    st_linestring(rbind(p1, p2))
  })
  st_sf(df, geometry = st_sfc(lines, crs = 4326))
}

manta_sf <- make_tow_lines(manta_raw)
manta_sf <- st_transform(manta_sf, crs = crs(prob_year2025))
manta_selected <- manta_sf %>% filter(ReefName %in% reef_select)

# Load Cull Data
cull_raw <- read_excel(cull_xlsx, sheet = 4) %>%
  filter(!is.na(Longitude), !is.na(Latitude), !is.na(Bottomtime), Bottomtime > 0, Longitude
  filter(as.Date(SurveyDate) > as.Date("2018-11-30")) %>%
  mutate(
    Total = Cohort1 + Cohort2 + Cohort3 + Cohort4,
    CPUE = Total / Bottomtime,
```

```

    is_ghost = (Total == 0),
    is_non_problematic = (CPUE < 0.02),
    ReefName = as.character(ReefName)
  )

# Top 30 reefs with most culling hours in 2025
reef_select_eval <- cull_raw %>%
  filter(as.Date(SurveyDate) >= as.Date("2025-01-01")) %>%
  group_by(ReefName) %>%
  summarise(Total_Hours = sum(Bottomtime, na.rm=TRUE) / 60) %>%
  arrange(desc(Total_Hours)) %>%
  slice_head(n = 30) %>%
  pull(ReefName)

# Manta tows for the selected 10 mapping reefs AND the 30 evaluated reefs
manta_selected <- manta_sf %>% filter(ReefName %in% reef_select)
manta_eval_all <- manta_sf %>% filter(ReefName %in% reef_select_eval)

# Cull dives for the selected 10 mapping reefs
cull_selected <- cull_raw %>% filter(ReefName %in% reef_select)
cull_sf <- st_as_sf(cull_selected, coords = c("Longitude", "Latitude"), crs = 4326) %>%
  st_transform(crs = crs(prob_year2025))

# Cull dives for the 30 evaluated reefs (point geometry)
cull_sf_eval_all <- st_as_sf(cull_raw %>% filter(ReefName %in% reef_select_eval), coords = c("Longitude", "Latitude"), crs = 4326) %>%
  st_transform(crs = crs(prob_year2025))

# Load cull site polygons and join to cull data
cull_site_polys <- st_read(cull_polys_gpkg, quiet = TRUE) %>%
  st_transform(crs = crs(prob_year2025))

cat("Loaded", nrow(cull_site_polys), "cull site polygons from gpkg\n")

```

Loaded 15884 cull site polygons from gpkg

```
cat("Matching to", n_distinct(cull_raw$CullSiteName), "unique cull site names...\n")
```

Matching to 3729 unique cull site names...

```
cat("Match rate:", round(mean(cull_raw$CullSiteName %in% cull_site_polys$CullSiteName) * 100
```

Match rate: 100 %

```
# Calculate Default Threshold: Median of 2025 raster cropped to selected reefs
ext_val <- terra::ext(manta_selected) + 2000
crop_2025 <- terra::crop(prob_year2025, ext_val)
```

```
|-----|-----|-----|-----|
=====
```

```
median_thresh_2025 <- as.numeric(terra::global(crop_2025, fun=function(x) median(x, na.rm=TRUE)
crop_agnostic <- terra::crop(prob_agnostic, ext_val)
```

```
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=====
```

```
median_thresh_agnostic <- as.numeric(terra::global(crop_agnostic, fun=function(x) median(x, na.rm=TRUE)

# Calculate P88 and P95 Thresholds
thresh_2025_p88 <- as.numeric(terra::global(crop_2025, fun=function(x) quantile(x, probs=0.88)
thresh_2025_p95 <- as.numeric(terra::global(crop_2025, fun=function(x) quantile(x, probs=0.95)

# Fixed mapping cutoffs
p_min_2025 <- 0.48
p_max_2025 <- 0.55

cat("Median Threshold (2025):", round(median_thresh_2025, 3), "\n")
```

Median Threshold (2025): 0.484

```
cat("Median Threshold (Agnostic):", round(median_thresh_agnostic, 3), "\n")
```

Median Threshold (Agnostic): 0.562

```
cat("88th Percentile (2025):", round(thresh_2025_p88, 3), "\n")
```

88th Percentile (2025): 0.629

```
cat("95th Percentile (2025):", round(thresh_2025_p95, 3), "\n")
```

95th Percentile (2025): 0.683

```
cat("Mapping Range (2025):", p_min_2025, "to", p_max_2025, "\n")
```

Mapping Range (2025): 0.48 to 0.55

1. Model Fitting Performance

This section presents the internal performance metrics generated during the XGBoost model training process (`fit_predict_cots_sdm.R`). The model was trained using 5-fold Spatial Block Cross-Validation (50km blocks) on historical cull dive data.

- **Training (In-Fold)** represents the model's performance when tested on the data it was trained on.
- **Validation (Out-of-Fold)** represents the model's performance on the hold-out spatial blocks, providing a robust estimate of how the model generalizes to new reefs.

```
metrics_file <- "outputs/model_performance_metrics.rds"

if (file.exists(metrics_file)) {
  metrics_list <- readRDS(metrics_file)

  # Overall Metrics Table
  cat("### 1.1 Overall Performance: Training vs Validation\n\n")
  print(kable(metrics_list$Overall, digits = 3, caption = "Overall Model Performance Metrics"))
  cat("\n\n")

  # Regional Metrics Table
  cat("### 1.2 Regional Validation Performance (Out-of-Fold)\n\n")
  print(kable(metrics_list$By_Zone %>% arrange(ManagementZone), digits = 3, caption = "Validation Performance by Management Zone"))
  cat("\n\n")

  # Spatial Block Variance Table
```

```

cat("### 1.3 Spatial Block Variance (Out-of-Fold)\n\n")
print(kable(metrics_list$By_Fold %>% arrange(id), digits = 3, caption = "Validation Performance Metrics")
cat("\n\n")

} else {
  cat("*Model performance metrics file not found. Ensure `fit_predict_cots_sdm.R` has been run.")
}

```

1.1 Overall Performance: Training vs Validation

Table 1: Overall Model Performance Metrics

Split	accuracy	sens	spec	roc_auc
Training (In-Fold)	0.792	0.890	0.676	0.125
Validation (Out-of-Fold)	0.592	0.762	0.388	0.390

1.2 Regional Validation Performance (Out-of-Fold)

Table 2: Validation Performance by Management Zone

ManagementZone	accuracy	sens	spec	roc_auc
Central	0.547	0.801	0.266	0.416
Far Northern	0.461	0.596	0.081	0.722
Northern	0.641	0.706	0.564	0.322
Southern	0.576	0.818	0.267	0.437

1.3 Spatial Block Variance (Out-of-Fold)

Table 3: Validation Performance by Spatial CV Fold

id	accuracy	sens	spec	roc_auc
Resample1	0.607	0.813	0.393	0.357
Resample2	0.525	0.791	0.234	0.467
Resample3	0.607	0.790	0.339	0.401
Resample4	0.632	0.639	0.622	0.320
Resample5	0.580	0.804	0.335	0.379

2. Step A: Grid-Based Historical Validation (100m)

We rasterize historical manta tow observations onto a 100m grid and compare them to the SDM probability values (using both 2025 and Agnostic models). We calculate ROC curves, AUC, and evaluate thresholds using confusion matrices.

```
if(nrow(manta_selected) > 0) {
  # crop_2025 is already 60m. Resample crop_agnostic to match it exactly.
  pred_60m_2025 <- crop_2025
  pred_60m_agn <- terra::resample(crop_agnostic, crop_2025, method = "bilinear")

  # Rasterize Manta Tow presence (1 if ever present, 0 if absent)
  manta_presence <- terra::rasterize(
    terra::vect(manta_selected),
    pred_60m_2025,
    field = "presence_any",
    fun = "max",
    background = NA
  )

  # Build evaluation dataframe
  val_df <- data.frame(
    observed_presence = terra::values(manta_presence)[,1],
    prob_2025 = terra::values(pred_60m_2025)[,1],
    prob_agnostic = terra::values(pred_60m_agn)[,1]
  ) %>% drop_na()

  # --- ROC and AUC ---
  roc_2025 <- roc(val_df$observed_presence, val_df$prob_2025, quiet=TRUE)
  roc_agn <- roc(val_df$observed_presence, val_df$prob_agnostic, quiet=TRUE)

  cat("AUC (2025 Model):", round(auc(roc_2025), 3), "\n")
  cat("AUC (Agnostic Model):", round(auc(roc_agn), 3), "\n\n")

  # Plot ROC
  par(mfrow=c(1,2))
  plot(roc_2025, main="ROC Curve (2025)", col="blue", lwd=2)
  plot(roc_agn, main="ROC Curve (Agnostic)", col="red", lwd=2)

  # --- Confusion Matrix at Median Threshold ---
  cat("Confusion Matrix for 2025 Model (Threshold = Median:", round(median_thresh_2025,3), ")")
  cm_2025 <- table(
    Predicted = factor(val_df$prob_2025 >= median_thresh_2025, levels=c(FALSE, TRUE)),

```



```

    Observed = factor(val_df$observed_presence == 1, levels=c(FALSE, TRUE))
  )
  print(cm_2025)
  cat("Accuracy:", sum(diag(cm_2025))/sum(cm_2025), "\n\n")

  cat("Confusion Matrix for Agnostic Model (Threshold = Median:", round(median_thresh_agnostic), ")")
  cm_agn <- table(
    Predicted = factor(val_df$prob_agnostic >= median_thresh_agnostic, levels=c(FALSE, TRUE)),
    Observed = factor(val_df$observed_presence == 1, levels=c(FALSE, TRUE))
  )
  print(cm_agn)
  cat("Accuracy:", sum(diag(cm_agn))/sum(cm_agn), "\n")
}

```

```

|-----|-----|-----|-----|
=====

```

```

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=====

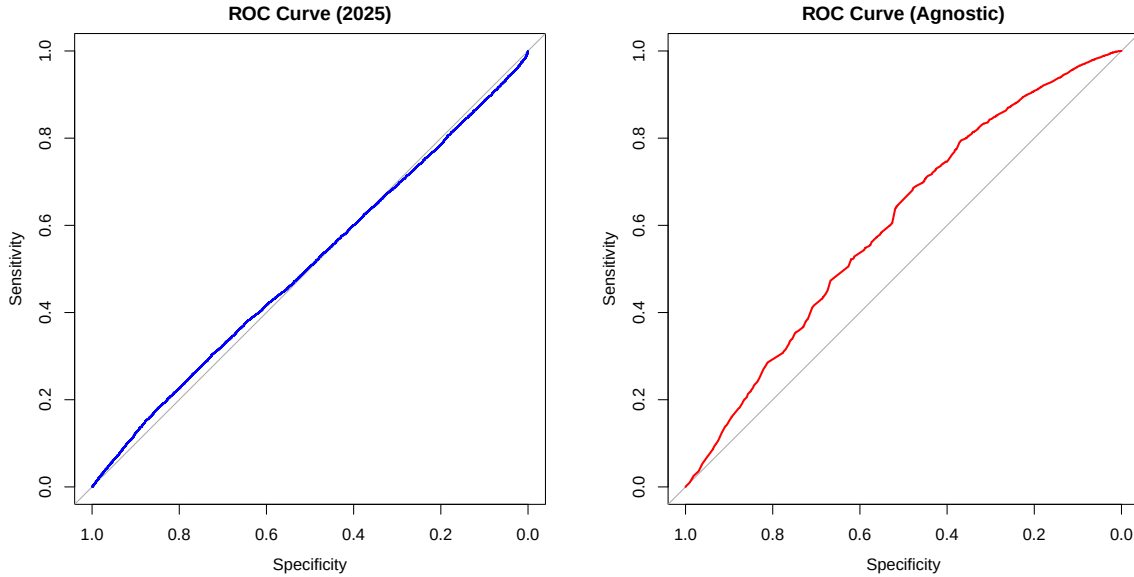
```

```

|-----|-----|-----|-----|
=====

```

AUC (2025 Model): 0.506
AUC (Agnostic Model): 0.602



Confusion Matrix for 2025 Model (Threshold = Median: 0.484)

```

Observed
Predicted FALSE TRUE
      FALSE 50771 4513
      TRUE  37710 3316
Accuracy: 0.5615928

```

Confusion Matrix for Agnostic Model (Threshold = Median: 0.562)

```

Observed
Predicted FALSE TRUE
      FALSE 60085 6499
      TRUE  28396 1330
Accuracy: 0.6376804

```

2b. Cull Dive Validation (ROC & AUC)

Cull dive data provides a more reliable validation target than manta tow data because divers directly count COTS at each site. We extract the SDM probability at each cull dive location and evaluate model discrimination using ROC curves. Presence is defined as `Total > 0` (at least one COTS found). We use all cull dive data from Jan 2025 onwards for the top 30 reefs.

```

# Extract SDM probabilities at cull dive point locations
cull_pts_2025 <- st_as_sf(
  cull_raw %>% filter(ReefName %in% reef_select_eval, as.Date(SurveyDate) >= as.Date("2025-0

```

```

  coords = c("Longitude", "Latitude"), crs = 4326
) %>% st_transform(crs = crs(prob_year2025))

# Extract 2025 model probability
cull_extract_2025 <- terra::extract(prob_year2025, terra::vect(cull_pts_2025))
val_col_2025 <- setdiff(names(cull_extract_2025), "ID")[1]

# Extract agnostic model probability
cull_extract_agn <- terra::extract(prob_agnostic, terra::vect(cull_pts_2025))
val_col_agn <- setdiff(names(cull_extract_agn), "ID")[1]

# Build validation dataframe
cull_val_df <- data.frame(
  observed_presence = as.integer(cull_pts_2025$Total > 0),
  prob_2025 = cull_extract_2025[[val_col_2025]],
  prob_agnostic = cull_extract_agn[[val_col_agn]],
  Total = cull_pts_2025$Total,
  CPUE = cull_pts_2025$CPUE
) %>% drop_na()

cat("Cull dive validation dataset:", nrow(cull_val_df), "dives\n")

```

Cull dive validation dataset: 1656 dives

```

cat("COTS present:", sum(cull_val_df$observed_presence),
    "(", round(mean(cull_val_df$observed_presence) * 100, 1), "%)\n")

```

COTS present: 1480 (89.4 %)

```

cat("COTS absent:", sum(!cull_val_df$observed_presence),
    "(", round(mean(!cull_val_df$observed_presence) * 100, 1), "%)\n\n")

```

COTS absent: 176 (10.6 %)

```

# --- ROC Curves ---
roc_cull_2025 <- roc(cull_val_df$observed_presence, cull_val_df$prob_2025, quiet = TRUE)
roc_cull_agn <- roc(cull_val_df$observed_presence, cull_val_df$prob_agnostic, quiet = TRUE)

cat("AUC (2025 Model vs Cull Data):", round(auc(roc_cull_2025), 3), "\n")

```

AUC (2025 Model vs Cull Data): 0.774

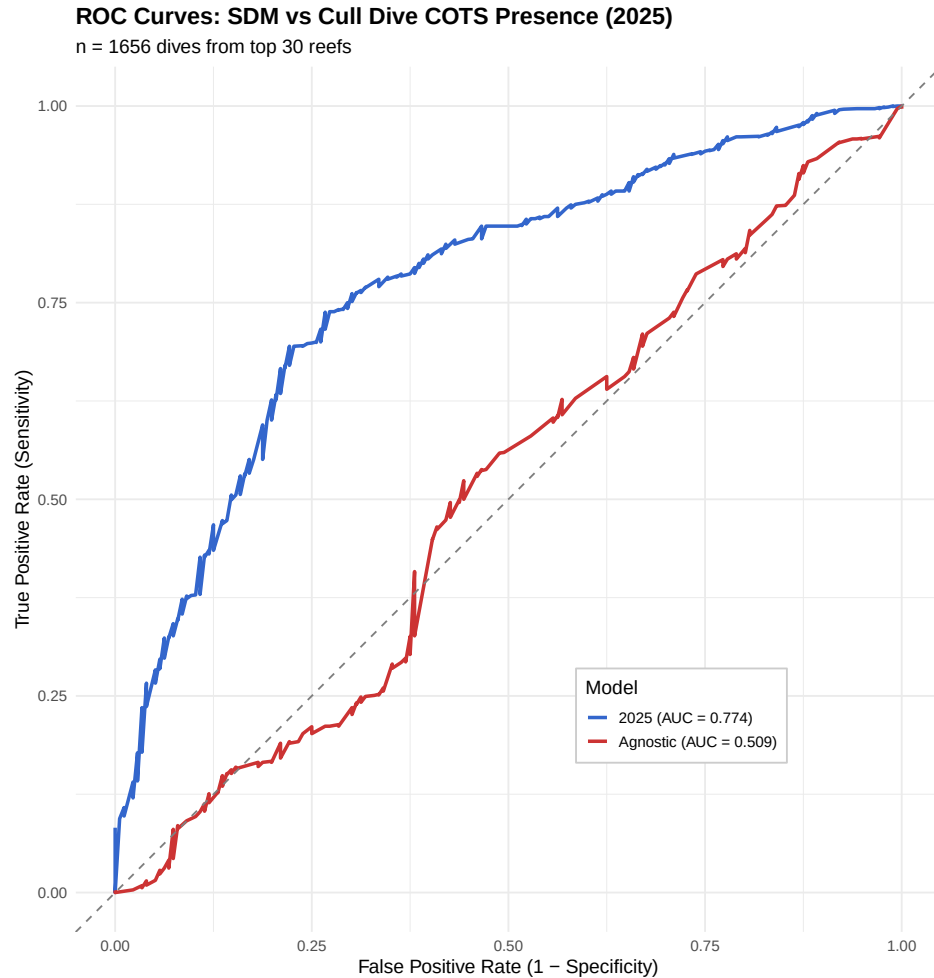
```
cat("AUC (Agnostic Model vs Cull Data):", round(auc(roc_cull_agn), 3), "\n\n")
```

AUC (Agnostic Model vs Cull Data): 0.509

```
# Build ROC dataframes for ggplot
roc_df_2025 <- data.frame(
  FPR = 1 - roc_cull_2025$specificities,
  TPR = roc_cull_2025$sensitivities,
  Model = paste0("2025 (AUC = ", round(auc(roc_cull_2025), 3), ")")
)
roc_df_agn <- data.frame(
  FPR = 1 - roc_cull_agn$specificities,
  TPR = roc_cull_agn$sensitivities,
  Model = paste0("Agnostic (AUC = ", round(auc(roc_cull_agn), 3), ")")
)
roc_df <- bind_rows(roc_df_2025, roc_df_agn)

# ggplot ROC curves
p_roc <- ggplot(roc_df, aes(x = FPR, y = TPR, color = Model)) +
  geom_line(linewidth = 1.2) +
  geom_abline(linetype = "dashed", color = "grey50") +
  scale_color_manual(values = c("#3366cc", "#cc3333")) +
  theme_minimal(base_size = 14) +
  labs(
    title = "ROC Curves: SDM vs Cull Dive COTS Presence (2025)",
    subtitle = paste0("n = ", nrow(cull_val_df), " dives from top 30 reefs"),
    x = "False Positive Rate (1 - Specificity)",
    y = "True Positive Rate (Sensitivity)",
    color = "Model"
  ) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = c(0.7, 0.25),
    legend.background = element_rect(fill = "white", color = "grey80")
  ) +
  coord_equal()

print(p_roc)
```



```
# --- Optimal thresholds (Youden's J) ---
opt_2025 <- coords(roc_cull_2025, "best", best.method = "youden", ret = c("threshold", "sens", "spec"))
opt_agn <- coords(roc_cull_agn, "best", best.method = "youden", ret = c("threshold", "sens", "spec"))

cat("Optimal Threshold (Youden's J) for 2025 Model:", round(opt_2025$threshold, 4),
    " Sensitivity:", round(opt_2025$sensitivity, 3),
    " Specificity:", round(opt_2025$specificity, 3), "\n")
```

Optimal Threshold (Youden's J) for 2025 Model: 0.4629 Sensitivity: 0.695 Specificity: 0.7

```
cat("Optimal Threshold (Youden's J) for Agnostic Model:", round(opt_agn$threshold, 4),
    " Sensitivity:", round(opt_agn$sensitivity, 3),
    " Specificity:", round(opt_agn$specificity, 3), "\n\n")
```

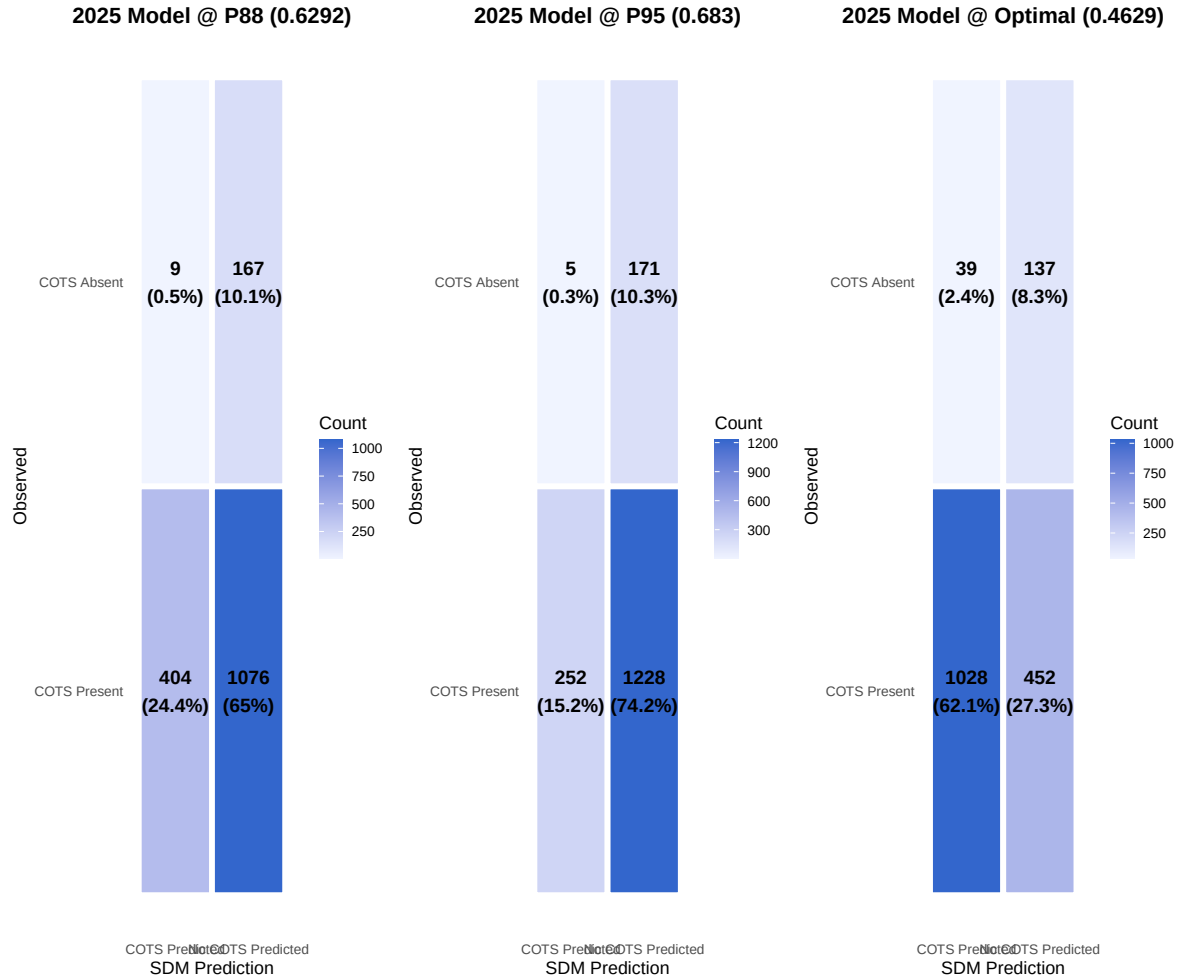
Optimal Threshold (Youden's J) for Agnostic Model: 0.5227 Sensitivity: 0.524 Specificity

```
# --- Confusion Matrices at P88 and P95 thresholds ---
plot_cull_cm <- function(prob_vec, obs_vec, threshold, title) {
  cm_df <- data.frame(
    Predicted = factor(ifelse(prob_vec >= threshold, "COTS Predicted", "No COTS Predicted"),
                        levels = c("COTS Predicted", "No COTS Predicted")),
    Observed = factor(ifelse(obs_vec, "COTS Present", "COTS Absent"),
                      levels = c("COTS Present", "COTS Absent"))
  )
  cm_counts <- cm_df %>%
    count(Predicted, Observed) %>%
    group_by() %>%
    mutate(pct = n / sum(n) * 100)

  ggplot(cm_counts, aes(x = Predicted, y = Observed, fill = n)) +
    geom_tile(color = "white", linewidth = 2) +
    geom_text(aes(label = paste0(n, "\n(", round(pct, 1), "%)")), size = 5, fontface = "bold") +
    scale_fill_gradient(low = "#f0f4ff", high = "#3366cc") +
    theme_minimal(base_size = 13) +
    labs(title = title, x = "SDM Prediction", y = "Observed", fill = "Count") +
    theme(plot.title = element_text(face = "bold", hjust = 0.5), panel.grid = element_blank())
}

p_cm1 <- plot_cull_cm(cull_val_df$prob_2025, cull_val_df$observed_presence == 1,
                     thresh_2025_p88, paste0("2025 Model @ P88 (", round(thresh_2025_p88, 4), "%)"))
p_cm2 <- plot_cull_cm(cull_val_df$prob_2025, cull_val_df$observed_presence == 1,
                     thresh_2025_p95, paste0("2025 Model @ P95 (", round(thresh_2025_p95, 4), "%)"))
p_cm3 <- plot_cull_cm(cull_val_df$prob_2025, cull_val_df$observed_presence == 1,
                     opt_2025$threshold, paste0("2025 Model @ Optimal (", round(opt_2025$threshold, 4), "%)"))

gridExtra::grid.arrange(p_cm1, p_cm2, p_cm3, ncol = 3)
```



3. Step B: Tow-Path Simulation and Evaluation

Simulating whether a specific tow path would have been suggested by the 2025 SDM model. We evaluate hit-rates strictly against tows conducted from Jan 1 2025 onwards, however, we use all historic tow tracks to delineate the operational boundaries.

```
# 2. Define Regions (100m Buffered Tows & 200m Grids)
tow_regions_list <- list()
grid_regions_list <- list()

for (r_name in unique(manta_eval_all$ReefName)) {
  m_sub <- manta_eval_all %>% filter(ReefName == r_name)
  if (nrow(m_sub) == 0) next
}
```

```

bb <- st_bbox(m_sub)
gr_500 <- st_make_grid(bb, cellsize = 500, square = TRUE) %>% st_as_sf()

# Tow Regions (100m buffer, unioned, then intersected with 500m grid)
tr <- st_buffer(m_sub, dist = 100) %>%
  st_union() %>%
  st_intersection(gr_500) %>%
  st_cast("POLYGON") %>%
  st_as_sf() %>%
  mutate(ReefName = r_name, RegionType = "Tow_100m_Seg500m", RegionID = paste0(r_name, "_T", row_number()),
         rename(geometry = x)
  tow_regions_list[[r_name]] <- tr

# Grid Regions (200m x 200m)
gr <- st_make_grid(bb, cellsize = 200, square = TRUE) %>%
  st_as_sf() %>%
  mutate(ReefName = r_name, RegionType = "Grid_200m", RegionID = paste0(r_name, "_G_", row_number()),
         rename(geometry = x)
  grid_regions_list[[r_name]] <- gr
}

tow_regions <- bind_rows(tow_regions_list)
grid_regions <- bind_rows(grid_regions_list)
all_regions <- bind_rows(tow_regions, grid_regions)

# 3. Extract SDM Stats for Regions
extract_poly_stats <- function(r, polys) {
  v <- terra::vect(polys)
  ex <- terra::extract(r, v)
  val_col <- setdiff(names(ex), "ID")[1]

  ex %>%
    rename(poly_id = ID, value = !!val_col) %>%
    group_by(poly_id) %>%
    summarise(
      reg_p88 = as.numeric(quantile(value, probs = 0.88, na.rm = TRUE, names = FALSE)),
      reg_p95 = as.numeric(quantile(value, probs = 0.95, na.rm = TRUE, names = FALSE)),
      .groups = "drop"
    )
}

```



```

    )
  }

all_regions$poly_id <- seq_len(nrow(all_regions))
region_stats <- extract_poly_stats(prob_year2025, all_regions)

all_regions <- all_regions %>% left_join(region_stats, by = "poly_id")
tow_regions <- all_regions %>% filter(RegionType == "Tow_100m_Seg500m")
grid_regions <- all_regions %>% filter(RegionType == "Grid_200m")

# 4. Map Tows to Regions and Apply Thresholds
# using st_join(largest = TRUE) ensures a tow is assigned to only one region
m_tow <- st_join(manta_eval_all, tow_regions %>% select(RegionID_Tow = RegionID, reg_p88_Tow = reg_p88_Tow))
m_grid <- st_join(manta_eval_all, grid_regions %>% select(RegionID_Grid = RegionID, reg_p88_Grid = reg_p88_Grid))

manta_eval <- manta_eval_all %>%
  mutate(
    reg_p88_Tow = m_tow$reg_p88_Tow,
    reg_p95_Tow = m_tow$reg_p95_Tow,
    reg_p88_Grid = m_grid$reg_p88_Grid,
    reg_p95_Grid = m_grid$reg_p95_Grid
  ) %>%
  mutate(
    sug_Tow_P88 = !is.na(reg_p88_Tow) & reg_p88_Tow >= thresh_2025_p88,
    sug_Tow_P95 = !is.na(reg_p95_Tow) & reg_p95_Tow >= thresh_2025_p95,
    sug_Grid_P88 = !is.na(reg_p88_Grid) & reg_p88_Grid >= thresh_2025_p88,
    sug_Grid_P95 = !is.na(reg_p95_Grid) & reg_p95_Grid >= thresh_2025_p95
  )

# Function to compute hit rates by reef for a specific scenario
calc_hit_rates_reef <- function(df, scenario_col, scenario_name) {
  overall <- df %>%
    summarise(
      ReefName = "Overall",
      Scenario = scenario_name,
      Total_Tows = n(),
      Tows_Suggested = sum(!is.na(scenario_col), na.rm=TRUE),
      Hit_Rate = sum(!is.na(scenario_col) & presence_any, na.rm=TRUE) / Tows_Suggested,
      Tows_Avoided = sum(!is.na(scenario_col), na.rm=TRUE),
      Miss_Rate = sum(!is.na(scenario_col) & presence_any, na.rm=TRUE) / Tows_Avoided
    )
  by_reef <- df %>%

```

```

group_by(ReefName) %>%
  summarise(
    Scenario = scenario_name,
    Total_Tows = n(),
    Tows_Suggested = sum(!sym(scenario_col), na.rm=TRUE),
    Hit_Rate = sum(!sym(scenario_col) & presence_any, na.rm=TRUE) / Tows_Suggested,
    Tows_Avoided = sum(! (sym(scenario_col)), na.rm=TRUE),
    Miss_Rate = sum(! (sym(scenario_col)) & presence_any, na.rm=TRUE) / Tows_Avoided
  )
bind_rows(by_reef, overall)
}

manta_eval_2025 <- manta_eval %>% filter(as.Date(SurveyTime) >= as.Date("2025-01-01"))

hit_table <- bind_rows(
  calc_hit_rates_reef(manta_eval_2025 %>% filter(ReefName %in% reef_select), "sug_Tow_P88", "P88"),
  calc_hit_rates_reef(manta_eval_2025 %>% filter(ReefName %in% reef_select), "sug_Tow_P95", "P95")
) %>% arrange(ReefName, desc(Scenario))

kable(hit_table, digits = 3, caption = "Manta Tow Simulation: Operational Zonal Validation (Selected 10 Reefs)")

```

Table 4: Manta Tow Simulation: Operational Zonal Validation (Selected 10 Reefs)

ReefName	Scenario	Total_Tows	Tows_Suggested	Hit_Rate	Tows_Avoided	Miss_Rate	Geometry
(Big) Broadhurst Reef (No 1) (18-100a)	Seg Tow 100m + P95	637	332	0.099	305	0.128	MULTILINESTRING ((573577.7 ...
(Big) Broadhurst Reef (No 1) (18-100a)	Seg Tow 100m + P88	637	185	0.076	452	0.128	MULTILINESTRING ((573577.7 ...
Batt Reef (16-029)	Seg Tow 100m + P95	382	344	0.308	38	0.026	MULTILINESTRING ((377707.6 ...
Batt Reef (16-029)	Seg Tow 100m + P88	382	317	0.312	65	0.123	MULTILINESTRING ((377707.6 ...
Davies Reef (18-096)	Seg Tow 100m + P95	177	100	0.200	77	0.260	MULTILINESTRING ((566203.3 ...

ReefName	Scenario	Total_Tows	Suggested_Tows	Hostile_Rate	Avoided_Tows	Miles_Ratio	Geometry
Davies Reef (18-096)	Seg Tow 100m + P88	177	60	0.117	117	0.282	MULTILINESTRING ((566203.3 ...
Heron Reef (23-052a)	Seg Tow 100m + P95	217	163	0.147	54	0.093	MULTILINESTRING ((1003623 7...
Heron Reef (23-052a)	Seg Tow 100m + P88	217	154	0.149	63	0.095	MULTILINESTRING ((1003623 7...
Overall	Seg Tow 100m + P95	1413	939	0.195	474	0.137	MULTILINESTRING ((377707.6 ...
Overall	Seg Tow 100m + P88	1413	716	0.200	697	0.151	MULTILINESTRING ((377707.6 ...

4. Step C: Cull Dive Financial Savings

Evaluating cull dives strictly from Jan 1 2025 onwards using the 2025 P88 and P95 thresholds on Segmented Tow Regions. We quantify avoided diving hours for dives that resulted in 0.00 CPUE (Ghost Culls) and <0.02 CPUE (Non-Problematic), and calculate financial savings (\$950/hr).

```
# --- Step 1: Build cull geometry using polygons where available, points otherwise ---
# Join cull data to polygon geometries by CullSiteName
cull_eval_raw <- cull_raw %>% filter(ReefName %in% reef_select_eval)
cull_eval_raw$row_id <- seq_len(nrow(cull_eval_raw))

# Match cull site names to polygon geometries
poly_lookup <- cull_site_polys %>% select(CullSiteName)

matched_idx <- cull_eval_raw$CullSiteName %in% poly_lookup$CullSiteName
cat("Cull site polygon match rate:", round(mean(matched_idx) * 100, 1), "%\n")
```

Cull site polygon match rate: 100 %

```
cat("  Matched:", sum(matched_idx), "  Unmatched:", sum(!matched_idx), "\n")
```

Matched: 9426 Unmatched: 0

```

# For matched: use polygon + 50m buffer
if (sum(matched_idx) > 0) {
  # Look up the polygon for each matched cull record
  match_positions <- match(cull_eval_raw$CullSiteName[matched_idx], poly_lookup$CullSiteName)
  matched_geoms <- st_buffer(st_geometry(poly_lookup)[match_positions], dist = 50)
  cull_matched_sf2 <- st_sf(cull_eval_raw[matched_idx, ], geometry = matched_geoms)
  cull_matched_sf2$match_type <- "Polygon"
} else {
  cull_matched_sf2 <- NULL
}

# For unmatched: use point + 100m buffer
if (sum(!matched_idx) > 0) {
  cull_unmatched_pts <- st_as_sf(cull_eval_raw[!matched_idx, ],
    coords = c("Longitude", "Latitude"), crs = 4326) %>%
    st_transform(crs = crs(prob_year2025))
  cull_unmatched_buf <- st_buffer(cull_unmatched_pts, dist = 100)
  cull_unmatched_buf$match_type <- "Point (100m buffer)"
} else {
  cull_unmatched_buf <- NULL
}

# Combine into single sf
cull_sf_combined <- bind_rows(cull_matched_sf2, cull_unmatched_buf) %>%
  arrange(row_id)

cat("Combined cull geometries:", nrow(cull_sf_combined), "rows\n")

```

Combined cull geometries: 9426 rows

```

# --- Step 2: Spatial join to tow/grid regions ---
c_tow <- st_join(cull_sf_combined, tow_regions %>% select(RegionID_Tow = RegionID, reg_p88))
c_grid <- st_join(cull_sf_combined, grid_regions %>% select(RegionID_Grid = RegionID, reg_p88))

# --- Step 3: Sector join (using centroids for sector assignment) ---
sectors_sf <- st_read("data/SectorShapefile/ltms.shp", quiet = TRUE) %>%
  st_transform(crs = crs(prob_year2025))
cull_centroids <- st_centroid(cull_sf_combined)
c_sector <- st_join(cull_centroids, sectors_sf %>% select(SECTOR_NAM), join = st_nearest_feature)

# --- Step 4: For unmatched tow-region NAs, extract MEAN raster value across the cull polygon

```

```
na_tow_idx <- is.na(c_tow$reg_p88_Tow)
cat("Tow region NA count:", sum(na_tow_idx), "of", nrow(c_tow), "\n")
```

Tow region NA count: 12 of 9426

```
# Extract mean raster value for each cull site polygon/buffer that didn't match a tow region
if (sum(na_tow_idx) > 0) {
  na_polys <- cull_sf_combined[na_tow_idx, ]
  na_extract <- terra::extract(prob_year2025, terra::vect(na_polys), fun = "mean", na.rm = TRUE)
  val_col <- setdiff(names(na_extract), "ID")[1]
  inferred_vals <- na_extract[[val_col]]
} else {
  inferred_vals <- numeric(0)
}

# --- Step 5: Build cull_eval with combined logic ---
cull_eval <- cull_sf_combined %>%
  st_drop_geometry() %>%
  mutate(
    reg_p88_Tow = c_tow$reg_p88_Tow,
    reg_p95_Tow = c_tow$reg_p95_Tow,
    reg_p88_Grid = c_grid$reg_p88_Grid,
    reg_p95_Grid = c_grid$reg_p95_Grid,
    SECTOR_NAM = c_sector$SECTOR_NAM
  )

# For NAs: fill in with inferred raster mean value
cull_eval$inferred_mean <- NA_real_
cull_eval$inferred_mean[na_tow_idx] <- inferred_vals

cull_eval <- cull_eval %>%
  mutate(
    # Region-matched classification
    avoid_Tow_P88 = case_when(
      !is.na(reg_p88_Tow) ~ reg_p88_Tow < thresh_2025_p88,
      TRUE ~ inferred_mean < thresh_2025_p88
    ),
    avoid_Tow_P95 = case_when(
      !is.na(reg_p95_Tow) ~ reg_p95_Tow < thresh_2025_p95,
      TRUE ~ inferred_mean < thresh_2025_p95
    ),
  )
```

```

avoid_Grid_P88 = case_when(
  !is.na(reg_p88_Grid) ~ reg_p88_Grid < thresh_2025_p88,
  TRUE ~ inferred_mean < thresh_2025_p88
),
avoid_Grid_P95 = case_when(
  !is.na(reg_p95_Grid) ~ reg_p95_Grid < thresh_2025_p95,
  TRUE ~ inferred_mean < thresh_2025_p95
),
# Transparency flag
classification_type = case_when(
  !is.na(reg_p88_Tow) ~ "Region Matched",
  !is.na(inferred_mean) ~ "Inferred",
  TRUE ~ "Unknown"
)
)

cat("\nClassification type breakdown:\n")

```

Classification type breakdown:

```
print(table(cull_eval$classification_type))
```

Inferred	Region Matched
12	9414

```
diver_hour_cost <- 950
```

```

calc_savings_reef <- function(df, avoid_col, scenario_name) {
  overall <- df %>%
    summarise(
      ReefName = "Overall",
      Scenario = scenario_name,
      Ghost_Avoided = sum(is_ghost & !!sym(avoid_col)),
      Ghost_Savings = sum(Bottomtime[is_ghost & !!sym(avoid_col)]) / 60 * diver_hour_cost,
      NonProb_Avoided = sum(is_non_problematic & !!sym(avoid_col)),
      NonProb_Savings = sum(Bottomtime[is_non_problematic & !!sym(avoid_col)]) / 60 * diver_
      Total_Savings = Ghost_Savings + NonProb_Savings,
      Prob_Missed = sum(!is_non_problematic & !!sym(avoid_col)),

```

```

    COTS_Missed = sum(Total[!is_non_problematic & !!sym(avoid_col)]),
    Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0),
    Hours_Missed = sum(Bottomtime[!is_non_problematic & !!sym(avoid_col)]) / 60
  ) %>%
  mutate(
    CPUE_Missed = ifelse(Hours_Missed > 0, COTS_Missed / (Hours_Missed * 60), 0)
  )
by_reef <- df %>%
  group_by(ReefName) %>%
  summarise(
    Scenario = scenario_name,
    Ghost_Avoided = sum(is_ghost & !!sym(avoid_col)),
    Ghost_Savings = sum(Bottomtime[is_ghost & !!sym(avoid_col)]) / 60 * diver_hour_cost,
    NonProb_Avoided = sum(is_non_problematic & !!sym(avoid_col)),
    NonProb_Savings = sum(Bottomtime[is_non_problematic & !!sym(avoid_col)]) / 60 * diver_hour_cost,
    Total_Savings = Ghost_Savings + NonProb_Savings,
    Prob_Missed = sum(!is_non_problematic & !!sym(avoid_col)),
    COTS_Missed = sum(Total[!is_non_problematic & !!sym(avoid_col)]),
    Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0),
    Hours_Missed = sum(Bottomtime[!is_non_problematic & !!sym(avoid_col)]) / 60
  ) %>%
  mutate(
    CPUE_Missed = ifelse(Hours_Missed > 0, COTS_Missed / (Hours_Missed * 60), 0)
  )
bind_rows(by_reef, overall)
}

cull_eval_2025 <- cull_eval %>%
  filter(as.Date(SurveyDate) >= as.Date("2025-01-01")) %>%
  mutate(
    SECTOR_NAM = as.character(SECTOR_NAM),
    ManagementZone = case_when(
      SECTOR_NAM %in% c("Cape Grenville", "Princess Charlotte Bay") ~ "Far Northern",
      SECTOR_NAM %in% c("Cooktown / Lizard Island", "Cairns", "Innisfail") ~ "Northern",
      SECTOR_NAM %in% c("Townsville", "Cape Upstart", "Whitsunday") ~ "Central",
      SECTOR_NAM %in% c("Pompey", "Swain", "Capricorn Bunker") ~ "Southern",
      TRUE ~ "Unknown"
    )
  )

# Only show reef-level summary for the 10 selected reefs to avoid massive tables
savings_table <- bind_rows(

```

```

    calc_savings_reef(cull_eval_2025 %>% filter(ReefName %in% reef_select), "avoid_Tow_P88", "
    calc_savings_reef(cull_eval_2025 %>% filter(ReefName %in% reef_select), "avoid_Tow_P95", "
  ) %>% arrange(ReefName, desc(Scenario))

kable(savings_table, digits = 2, format.args = list(big.mark = ","), caption = "Financial Sa

```

Table 5: Financial Savings and Risk Summary (Selected 10 Reefs)

ReefName	Scenario	Ghost	Avoided	Savings	Non-Pooled	AP	Pool	Savings	Base	AVG	SP	Missed	Costs	NPV	Missed
(Big)	Seg	27	103,439.1770		321,179.42	24,618.340		987	88.52	234.17	0.07				
Broadhurst	Tow														
Reef (No 1)	100m														
(18-100a)	+ P95														
(Big)	Seg	27	103,439.1770		321,179.42	24,618.340		987	88.52	234.17	0.07				
Broadhurst	Tow														
Reef (No 1)	100m														
(18-100a)	+ P88														
Batt Reef	Seg	4	23,591.67	21	119,700.04	3,291.626		4,349	98.73	804.43	0.09				
(16-029)	Tow														
	100m														
	+ P95														
Batt Reef	Seg	4	23,591.67	21	119,700.04	3,291.626		4,349	98.73	804.43	0.09				
(16-029)	Tow														
	100m														
	+ P88														
Davies Reef	Seg	3	9,341.67	22	72,105.08	1,446.6724		167	77.31	87.05	0.03				
(18-096)	Tow														
	100m														
	+ P95														
Davies Reef	Seg	3	9,341.67	22	72,105.08	1,446.6724		167	77.31	87.05	0.03				
(18-096)	Tow														
	100m														
	+ P88														
Heron Reef	Seg	9	33,281.67	47	259,270.29	2,552.513		214	63.50	87.48	0.04				
(23-052a)	Tow														
	100m														
	+ P95														
Heron Reef	Seg	9	33,281.67	47	259,270.29	2,552.513		214	63.50	87.48	0.04				
(23-052a)	Tow														
	100m														
	+ P88														

ReefName	Scenario	Ghost_Avoided	NonProb_Savings	Total_Savings	COTS_Missed	Hours_Missed
Overall	Seg Tow 100m + P95	43	169,654.1760	772,255.941,909.203	5,717 94.14	1,213.16.08
Overall	Seg Tow 100m + P88	43	169,654.1760	772,255.941,909.203	5,717 94.14	1,213.16.08

```
# Management Zone & Sector Summary (All 128 Reefs)
calc_savings_zone <- function(df, avoid_col, scenario_name) {
  df %>%
    group_by(ManagementZone, SECTOR_NAM) %>%
    summarise(
      Scenario = scenario_name,
      Ghost_Savings = sum(Bottomtime[is_ghost & !!sym(avoid_col)]) / 60 * diver_hour_cost,
      NonProb_Savings = sum(Bottomtime[is_non_problematic & !!sym(avoid_col)]) / 60 * diver_hour_cost,
      Total_Savings = Ghost_Savings + NonProb_Savings,
      COTS_Missed = sum(Total[!is_non_problematic & !!sym(avoid_col)]),
      Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0),
      Hours_Missed = sum(Bottomtime[!is_non_problematic & !!sym(avoid_col)]) / 60,
      .groups = "drop"
    )
}

savings_zone_table <- bind_rows(
  calc_savings_zone(cull_eval_2025, "avoid_Tow_P88", "Seg Tow 100m + P88"),
  calc_savings_zone(cull_eval_2025, "avoid_Tow_P95", "Seg Tow 100m + P95")
) %>%
  arrange(factor(ManagementZone, levels = c("Far Northern", "Northern", "Central", "Southern")))

kable(savings_zone_table, digits = 2, format.args = list(big.mark = ","), caption = "Financial Summary of Management Zones and Sectors")
```

Table 6: Financial Savings and Risk Summary by Management Zone & Sector (Top 30 Reefs, 2025)

Management Zone	SECTOR	NAME	Scenario	Ghost_Savings	Prob_Savings	Total_Savings	GOS	Risks	COTS	Home	Missed
Far North-ern	Cape Grenville	Seg Tow 100m + P95		9,468.33	98,451.67	107,920.00	820	92.34	217.52		
Far North-ern	Cape Grenville	Seg Tow 100m + P88		9,468.33	98,451.67	107,920.00	820	92.34	217.52		
Northern	Cairns	Seg Tow 100m + P95		125,558.33	336,585.00	462,143.33	5,437	97.54	1,123.05		
Northern	Cairns	Seg Tow 100m + P88		125,558.33	336,585.00	462,143.33	5,437	97.54	1,123.05		
Northern	Cooktown / Lizard Island	Seg Tow 100m + P95		1,725.83	229,551.67	231,277.50	9,147	98.43	1,590.58		
Northern	Cooktown / Lizard Island	Seg Tow 100m + P88		1,725.83	229,551.67	231,277.50	9,147	98.43	1,590.58		
Northern	Innisfail	Seg Tow 100m + P95		12,033.33	49,400.00	61,433.33	1,117	98.59	196.77		
Northern	Innisfail	Seg Tow 100m + P88		12,033.33	49,400.00	61,433.33	1,117	98.59	196.77		
Central	Cape Upstart	Seg Tow 100m + P95		106,922.50	60,122.50	267,045.00	136	81.93	44.37		
Central	Cape Upstart	Seg Tow 100m + P88		106,922.50	60,122.50	267,045.00	136	81.93	44.37		
Central	Townsville	Seg Tow 100m + P95		161,547.50	14,593.33	976,140.83	6,069	93.93	1,235.98		
Central	Townsville	Seg Tow 100m + P88		161,547.50	14,593.33	976,140.83	6,069	93.93	1,235.98		

Management Zone	SECTOR_NAME	Scenario	Ghost_Savings	NonProb_Savings	Total_Savings	COTS_Missed	Hit_Missed	Hit_Missed
Southern	Capricorn Bunker	Seg Tow 100m + P95	79,214.17	658,983.33	738,197.50	4,178	92.23	1,072.78
Southern	Capricorn Bunker	Seg Tow 100m + P88	79,214.17	658,983.33	738,197.50	4,178	92.23	1,072.78
Southern	Swain	Seg Tow 100m + P95	192,264.17	407,930.00	600,194.17	2,260	95.16	442.27
Southern	Swain	Seg Tow 100m + P88	192,264.17	407,930.00	600,194.17	2,260	95.16	442.27

```
# --- Consolidated Scenario Summary ---
scenario_summary <- function(manta_df, cull_df, tow_col, avoid_col, label) {
  # Manta tow summary
  m <- manta_df %>%
    summarise(
      Total_Tows = n(),
      Tows_Suggested = sum(!sym(tow_col), na.rm=TRUE),
      Tows_Avoided = Total_Tows - Tows_Suggested,
      Hit_Rate = sum(!sym(tow_col) & presence_any, na.rm=TRUE) / max(Tows_Suggested, 1),
      Miss_Rate = sum(!(!sym(tow_col)) & presence_any, na.rm=TRUE) / max(Tows_Avoided, 1)
    )
  # Cull dive summary
  c <- cull_df %>%
    summarise(
      Total_Dives = n(),
      Dives_Avoided = sum(!sym(avoid_col), na.rm=TRUE),
      Ghost_Savings = sum(Bottomtime[is_ghost & !sym(avoid_col)]) / 60 * diver_hour_cost,
      NonProb_Savings = sum(Bottomtime[is_non_problematic & !sym(avoid_col)]) / 60 * diver_hour_cost,
      Total_Savings = Ghost_Savings + NonProb_Savings,
      COTS_Missed = sum(Total[!is_non_problematic & !sym(avoid_col)]),
      Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0)
    )
  tibble(
    Scenario = label,
    Total_Tows = m$Total_Tows, Tows_Suggested = m$Tows_Suggested, Tows_Avoided = m$Tows_Avoided,
    Tow_Hit_Rate = m$Hit_Rate, Tow_Miss_Rate = m$Miss_Rate,
    Total_Dives = c$Total_Dives, Dives_Avoided = c$Dives_Avoided,

```

```

    Total_Savings = c$Total_Savings, COTS_Missed = c$COTS_Missed, Pct_COTS_Missed = c$Pct_COTS_Missed
  )
}

consolidated <- bind_rows(
  scenario_summary(manta_eval_2025, cull_eval_2025, "sug_Tow_P88", "avoid_Tow_P88", "P88 (Agg-
  scenario_summary(manta_eval_2025, cull_eval_2025, "sug_Tow_P95", "avoid_Tow_P95", "P95 (Con
)

kable(consolidated, digits = 2, format.args = list(big.mark = ","),
      caption = "Consolidated Scenario Comparison: Manta Tow + Cull Dive (All Top 30 Reefs, 2025)"
)

```

Table 7: Consolidated Scenario Comparison: Manta Tow + Cull Dive (All Top 30 Reefs, 2025)

Scenario	Total_Tows	Suggested_Tows	Avoided_Tows	Hit_Rate	Miss_Rate	Total_Dives	Avoided_Dives	Total_Savings	COTS_Missed	COTS_Missed
P88 (Aggressive)	7,757	4,090	3,667	0.15	0.16	1,658	1,658	3,444,352	29,164	95.87
P95 (Conservative)	7,757	5,140	2,617	0.16	0.16	1,658	1,658	3,444,352	29,164	95.87

Confusion Matrices

```

# Function to build and plot a confusion matrix
plot_confusion_matrix <- function(predicted_suggested, observed_presence, title) {
  cm_df <- data.frame(
    Predicted = factor(ifelse(predicted_suggested, "Suggested", "Avoided"), levels = c("Suggested", "Avoided")),
    Observed = factor(ifelse(observed_presence, "COTS Present", "COTS Absent"), levels = c("COTS Present", "COTS Absent"))
  )
  cm_counts <- cm_df %>%
    count(Predicted, Observed) %>%
    group_by() %>%
    mutate(pct = n / sum(n) * 100)

  ggplot(cm_counts, aes(x = Predicted, y = Observed, fill = n)) +
    geom_tile(color = "white", linewidth = 2) +
    geom_text(aes(label = paste0(n, "\n(", round(pct, 1), "%)")), size = 5, fontface = "bold") +
    scale_fill_gradient(low = "#f0f4ff", high = "#3366cc") +
    theme_minimal()
}

```

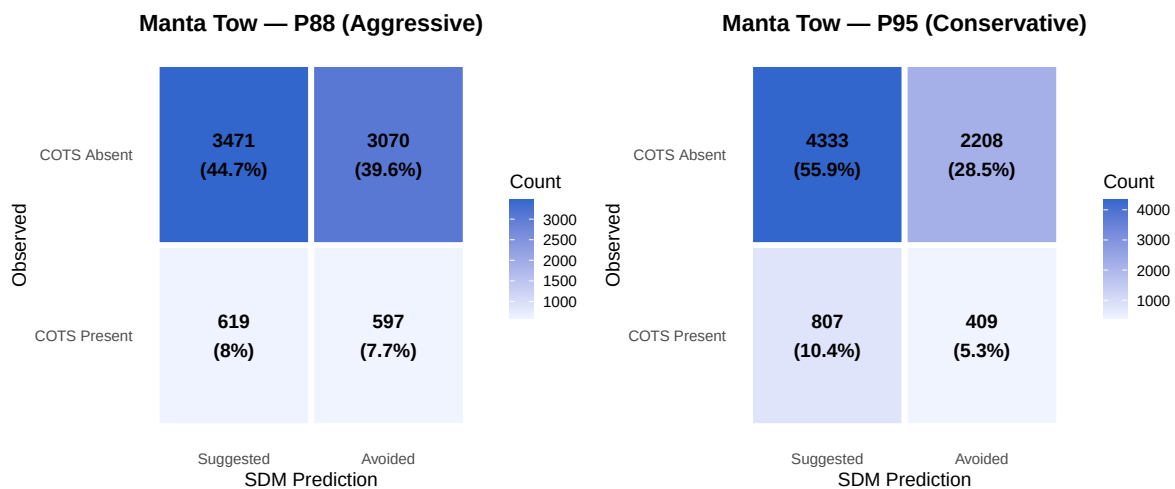
```

theme_minimal(base_size = 14) +
labs(title = title, x = "SDM Prediction", y = "Observed", fill = "Count") +
theme(
  plot.title = element_text(face = "bold", hjust = 0.5),
  panel.grid = element_blank()
)
}

# Manta Tow Confusion Matrices
p1 <- plot_confusion_matrix(
  manta_eval_2025$sug_Tow_P88, manta_eval_2025$presence_any,
  "Manta Tow - P88 (Aggressive)"
)
p2 <- plot_confusion_matrix(
  manta_eval_2025$sug_Tow_P95, manta_eval_2025$presence_any,
  "Manta Tow - P95 (Conservative)"
)

gridExtra::grid.arrange(p1, p2, ncol = 2)

```



```

# Cull Dive Confusion Matrices (is_non_problematic as "COTS Absent" proxy)
p3 <- plot_confusion_matrix(
  !cull_eval_2025$avoid_Tow_P88, !cull_eval_2025$is_non_problematic,
  "Cull Dive - P88 (Aggressive)"
)
p4 <- plot_confusion_matrix(
  !cull_eval_2025$avoid_Tow_P95, !cull_eval_2025$is_non_problematic,

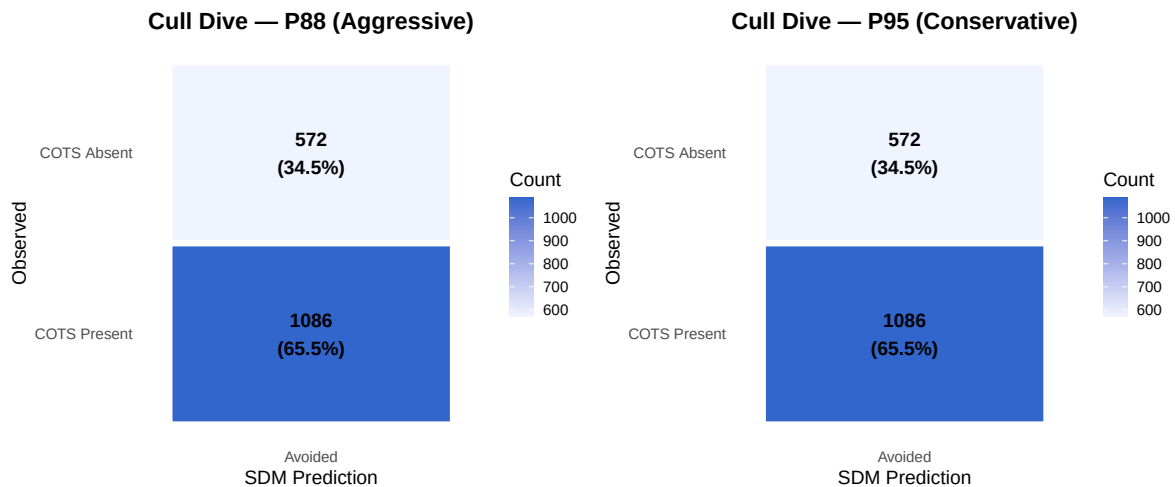
```

```

"Cull Dive - P95 (Conservative)"
)

gridExtra::grid.arrange(p3, p4, ncol = 2)

```



5. Step D: Visual Mapping

Below are maps for each reef showing the underlying 2025 SDM probability. The operational units displayed are the **100m Buffered Tow Regions**, categorized by the **95th Percentile (P95) Threshold** (Red = Suggested, Grey = Avoided). Individual manta tows and cull dive locations are also plotted to show what falls inside the suggested operational areas.

```

plot_reef_map <- function(r, tow_sf, dive_sf, region_sf, reef_name, p_min, p_max) {
  # Bounding box around features
  bb_t <- if(nrow(tow_sf) > 0) st_bbox(tow_sf) else NULL
  bb_d <- if(nrow(dive_sf) > 0) st_bbox(dive_sf) else NULL

  xmin <- min(c(bb_t["xmin"], bb_d["xmin"]), na.rm=TRUE)
  ymin <- min(c(bb_t["ymin"], bb_d["ymin"]), na.rm=TRUE)
  xmax <- max(c(bb_t["xmax"], bb_d["xmax"]), na.rm=TRUE)
  ymax <- max(c(bb_t["ymax"], bb_d["ymax"]), na.rm=TRUE)

  pad <- 2000 # 2km padding
  ext_map <- terra::ext(xmin - pad, xmax + pad, ymin - pad, ymax + pad)

  rc <- terra::crop(r, ext_map)
}

```

```

# Scale aggregate factor to keep cell count manageable (~10,000 pixels) for fast plotting
fact <- ceiling(sqrt(terra::ncell(rc) / 10000))
if(fact > 1) {
  rc <- terra::aggregate(rc, fact = fact, fun = "mean", na.rm = TRUE)
}

rdf <- as.data.frame(rc, xy = TRUE, na.rm = TRUE)
names(rdf)[3] <- "prob"

# Determine if region is suggested based on P95 threshold
region_sf <- region_sf %>%
  mutate(is_suggested = !is.na(reg_p95) & reg_p95 >= thresh_2025_p95)

gg <- ggplot() +
  geom_raster(data = rdf, aes(x = x, y = y, fill = prob)) +
  # Plot Operational Zones
  geom_sf(data = region_sf, aes(color = is_suggested), fill = NA, linewidth = 1.2, linetype = "dashed") +
  scale_color_manual(values = c("TRUE" = "red", "FALSE" = "grey70")) +
  # Plot Tows and Dives
  geom_sf(data = tow_sf, color = "black", linewidth = 0.5) +
  geom_sf(data = dive_sf, shape = 21, fill = "white", color = "black", size = 2) +
  scale_fill_viridis_c(
    option = "C",
    na.value = NA,
    limits = c(p_min, p_max),
    oob = scales::squish
  ) +
  theme_minimal() +
  labs(
    title = paste("Operational Zonal Map:", reef_name),
    subtitle = paste0("Red Dashed = Suggested 100m Zone (P95), Grey Dashed = Avoided Zone\\n", reef_name),
    fill = "2025 SDM Prob",
    color = "Suggested Zone"
  )
return(gg)
}

# Generate plots for each reef
cull_sf_2025 <- cull_sf %>% filter(as.Date(SurveyDate) >= as.Date("2025-01-01"))

for(r_name in reef_select) {
  t_sf <- manta_eval_2025 %>% filter(ReefName == r_name)
}

```

```

d_sf <- cull_sf_2025 %>% filter(ReefName == r_name)
r_sf <- tow_regions %>% filter(ReefName == r_name)

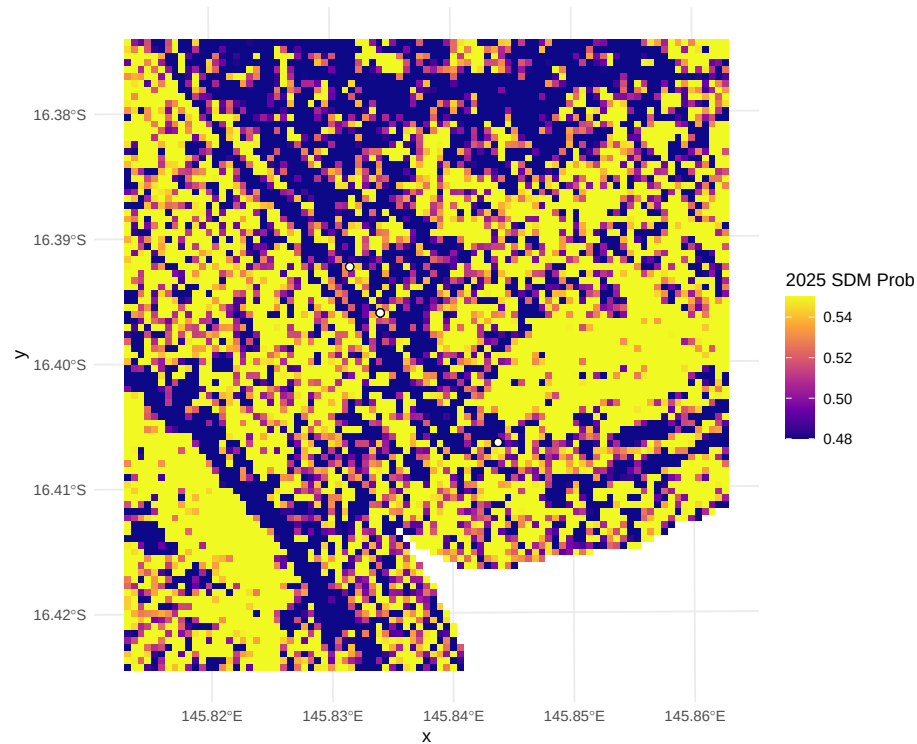
if(nrow(t_sf) > 0 || nrow(d_sf) > 0) {
  message(paste("Processing map for:", r_name))
  cat("\n### ", r_name, "\n")
  p <- plot_reef_map(prob_year2025, t_sf, d_sf, r_sf, r_name, p_min_2025, p_max_2025)
  print(p)
  cat("\n")
}
}

```

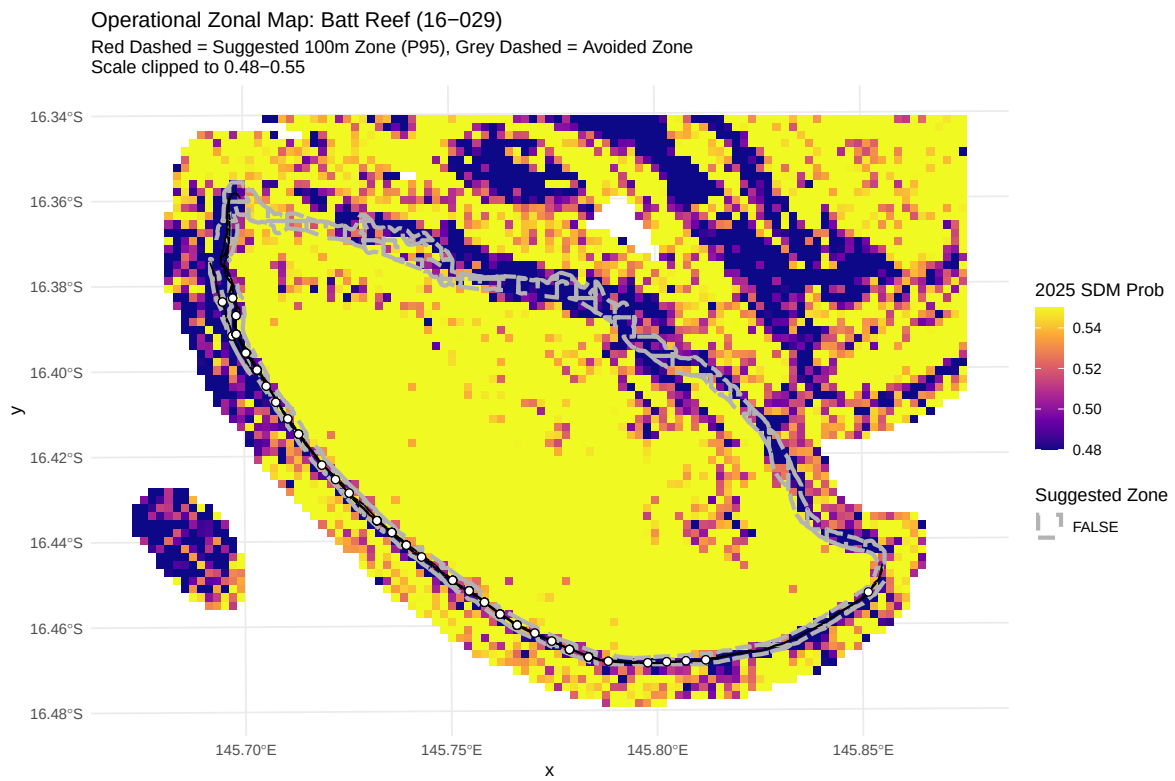
Tongue Reef (16-026)

Operational Zonal Map: Tongue Reef (16-026)

Red Dashed = Suggested 100m Zone (P95), Grey Dashed = Avoided Zone
Scale clipped to 0.48–0.55

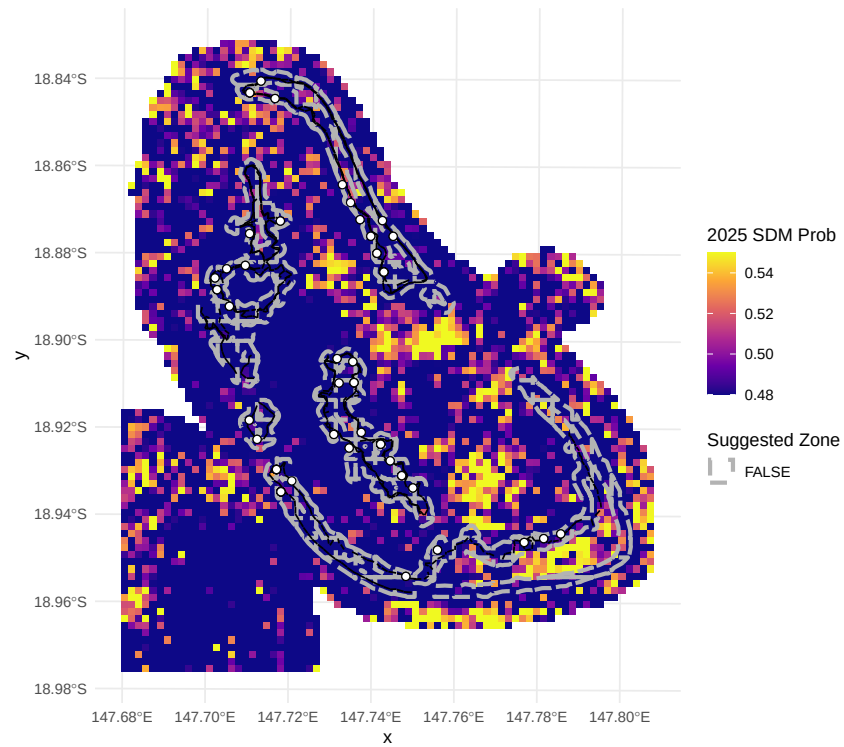


Batt Reef (16-029)



(Big) Broadhurst Reef (No 1) (18-100a)

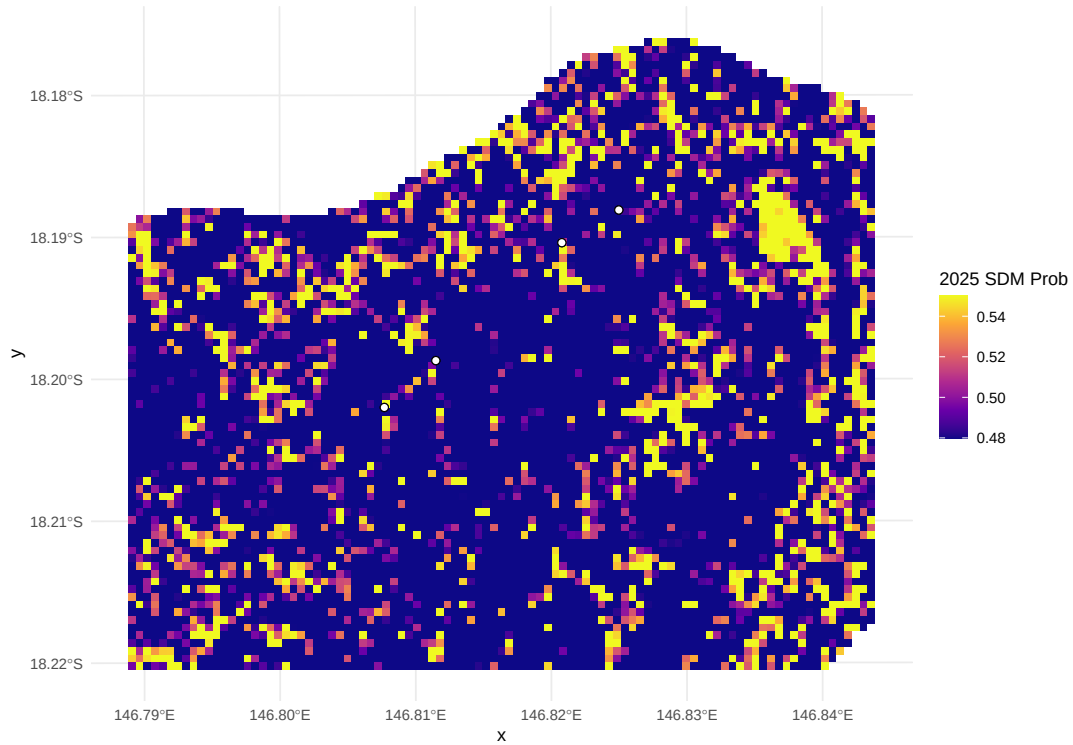
Operational Zonal Map: (Big) Broadhurst Reef (No 1) (18-100a)
Red Dashed = Suggested 100m Zone (P95), Grey Dashed = Avoided Zone
Scale clipped to 0.48-0.55



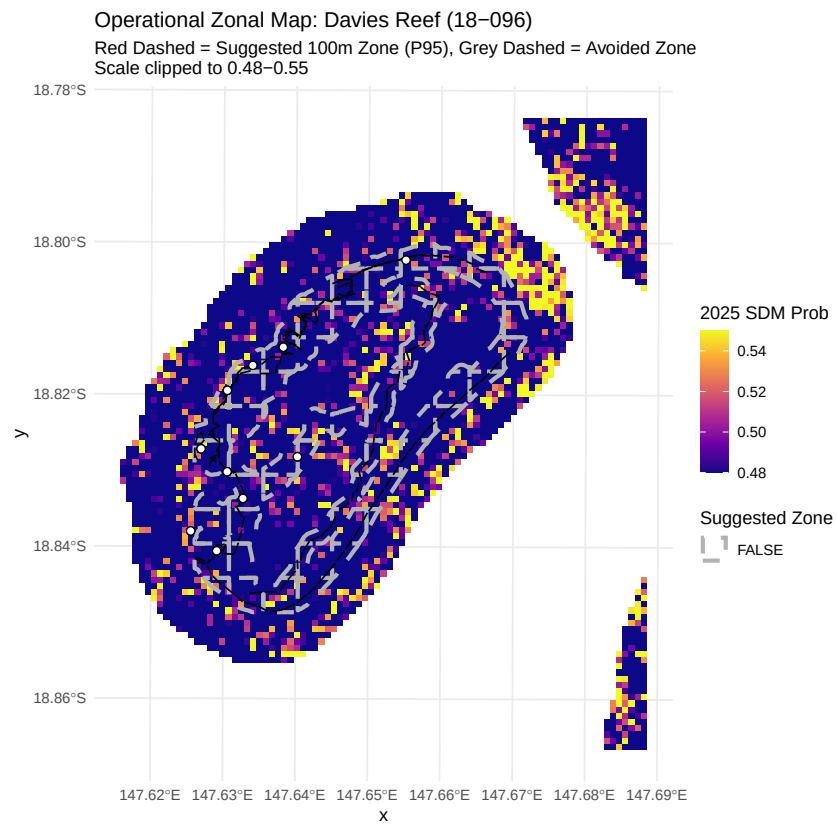
Britomart Reef (18-024)

Operational Zonal Map: Britomart Reef (18-024)

Red Dashed = Suggested 100m Zone (P95), Grey Dashed = Avoided Zone
Scale clipped to 0.48-0.55



Davies Reef (18-096)



Heron Reef (23-052a)

